



Fig. 2: Types of technologies involved in an IoT application

sensors directly communicating to the central server or a communication gateway.

Next to Things are Gateways, which manage multiple devices (Things) that cannot communicate with the central server. These gateways act like proxies to the central server for the devices these manage. Gateways can also handle multiple sensors in addition to gateway function.

Then comes the Internet, which is the connectivity conduit. The Internet essentially connects Things to the central server. This enables data from Things to be transferred to the central server or cloud (transfer of data from the server to Things is also possible). This connectivity conduit can be either wireline or wireless. Telecom/datacom service providers typically provide connectivity and mostly IP (IPv6) based networks. Today, most of these manage and provide end-to-end connections.

Finally, on top are central servers, which can be stand-alone servers or cloud based solutions depending on the business model and cost. These offer IoT services by running the applications centrally. However, the key is that bulk of the IoT value is derived out of central services and Things mostly act like agents that collect data and send it forward.

Fig. 1 also shows the type of communication technology that is used between layers. This essentially

means that Things can be an embedded system, which can either have its own communication capability, or can be a simple device that can be managed by a gateway, which will manage these devices.

While we look at the IoT architecture as a whole, we also need to remember that the ecosystem needs three different participants. While the IoT operates on the basic premises that use IPv6 protocol, Things can be designed without these supporting IPv6 and instead communicate with a different protocol to Gateways, which typically support IPv6 and forward data from Things to the central application.

The three participants in an IoT application are:

Users. Who use Things

Service providers. Who provide connectivity between Things and IoT solution providers

Solution providers. Who, through Things, Gateways and service providers, offer the final solution running from the central servers

To explain this, let us take the case of personal fitness application services provided by solution providers. The user buys the wearable (like a band) and other equipment, which measure different fitness parameters. These bands could be communicating with a smartphone either through Bluetooth or near field communication (NFC).

The smartphone has applications for communicating with Things (bands) as well as with the central application server for uploading data collected from Things.

Application service is a cloud-mounted fitness program customised for individual users. It collects data, analyses it and recommends the course correction for maintaining fitness. This essentially means a whole lot of technologies are used. Fig. 2 shows the types of technologies that are involved, for easy understanding.

Characteristics of IoT systems

Having seen some generic features of the IoT, let us now see some unique characteristics of IoT systems, which also impact the design of Things. IoT Things are classified into two major categories of devices, based on their capability and ability to communicate.

The first class of devices handles sensors but can not communicate with the central application server. Data from these devices can be sent to the central server only through Gateways. Typically, these devices use ZigBee, Bluetooth, NFC and RFID for communication. Primary need for such devices is that, these handle single sensors and data volume is

Classification of Constrained Devices

CLASS	RAM	Flash	Remarks
Class 0	<1kB	<100kB	These devices use Gateways to communicate and have basic communication capabilities only
Class 1	~10kB	~100kB	These devices use protocols specifically designed for IoT devices using Constrained Application Protocols (CoAP) and can interact with other devices without Gateways
Class 2	~50kB	~250kB	These devices support regular IP (IPv6) protocols and can behave like other network devices